WORKED SOLUTION

STEP I 1998, Question 4

Pure • Constrained Optimisation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q4

The key idea

A rectangle inscribed in a circle has its diagonals as diameters, so its sides a and b are tied together by Pythagoras: $a^2 + b^2$ is fixed. Everything then becomes a question about how $a + b$, $a^2 + b^2$ and $a^3 + b^3$ behave when $a^2 + b^2$ is held constant. The case $n = 2$ is the hinge: below it the square wins, at it everything ties, and above it the square comes last.

Solution

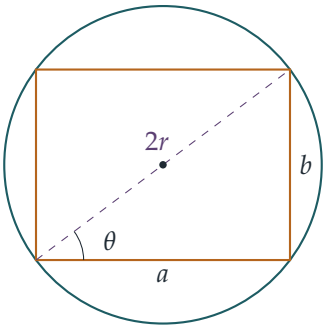
Let the circle have radius r and let the rectangle have sides $a, b > 0$. Each angle of the rectangle is a right angle, so it subtends a diameter (the angle in a semicircle); hence each diagonal is a diameter and

$$a^2 + b^2 = (2r)^2 = 4r^2.$$

Conversely, any $a, b > 0$ with $a^2 + b^2 = 4r^2$ give a rectangle that fits in the circle with its vertices on it. So it is convenient to write

$$a = 2r\cos\theta, \qquad b = 2r\sin\theta, \qquad 0 < \theta < \tfrac{\pi}{2},$$

where θ is the angle between a diagonal and a side of length a.



The perimeter

The perimeter is $P = 2(a+b)$. Since $(a-b)^2 \geq 0$, we have $2ab \leq a^2 + b^2$, so

$$(a+b)^2 = a^2 + b^2 + 2ab \leq 2(a^2+b^2) = 8r^2,$$

with equality if and only if $a = b$. As $a + b > 0$, this gives $a + b \leq 2\sqrt{2}\,r$, that is

$$P \leq 4\sqrt{2}\,r,$$

and the value $4\sqrt{2}\,r$ is attained exactly when $a = b = \sqrt{2}\,r$. So the rectangle of greatest perimeter is the square, as required.

The case $n = 2$

The sum of the squares of the four sides is

$$2(a^2+b^2) = 8r^2,$$

whatever the rectangle. So every inscribed rectangle gives the same value: there is nothing to choose between them, and in particular the square is no better than any other.

The case $n = 3$

Now we look at $S = 2(a^3+b^3)$. Since each side is shorter than the diameter, $0 < a < 2r$ and $0 < b < 2r$, so $a^3 < 2r\,a^2$ and $b^3 < 2r\,b^2$. Adding,

$$S = 2(a^3+b^3) < 2 \cdot 2r(a^2+b^2) = 16r^3.$$

On the other hand, as $\theta \to 0$ (a long thin rectangle, with $a \to 2r$ and $b \to 0$), $S \to 2(2r)^3 = 16r^3$. So S can be made as close to $16r^3$ as we like but never reaches it: *there is no rectangle of greatest S*. The only "shape" achieving $16r^3$ is the degenerate one, a diameter traced twice, which is not a rectangle.

? **Ask yourself** So what has happened to the square?

It has gone from best to worst. With $f(\theta) = \cos^3\theta + \sin^3\theta$, we have $S = 16r^3 f(\theta)$ and

$$f'(\theta) = -3\cos^2\theta\sin\theta + 3\sin^2\theta\cos\theta = 3\sin\theta\cos\theta\,(\sin\theta - \cos\theta).$$

On $0 < \theta < \frac{\pi}{2}$ the factor $\sin\theta\cos\theta$ is positive, and $\sin\theta - \cos\theta$ is negative for $\theta < \frac{\pi}{4}$ and positive for $\theta > \frac{\pi}{4}$. So f decreases and then increases, and the square ($\theta = \frac{\pi}{4}$) gives the *least* value,

$$S_{\min} = 16r^3\left(2 \cdot \frac{1}{2\sqrt{2}}\right) = 8\sqrt{2}\,r^3.$$

In summary, for $n = 3$ the values of S fill the interval $[8\sqrt{2}\,r^3,\ 16r^3)$: the square minimises, and no rectangle maximises.

The same happens for every $n > 2$: $\cos^n\theta + \sin^n\theta < \cos^2\theta + \sin^2\theta = 1$ on $0 < \theta < \frac{\pi}{2}$, because both $\cos\theta$ and $\sin\theta$ lie strictly between 0 and 1, so the supremum $2(2r)^n$ is approached only by flattening the rectangle.

Checks

- With $r = 1.7$ we scanned θ over $(0, \frac{\pi}{2})$ numerically: the perimeter peaks at $\theta = \frac{\pi}{4}$ with value $4\sqrt{2}\,r \approx 9.617$; the sum of squares is constantly $8r^2 = 23.12$; and the sum of cubes falls from near $16r^3 \approx 78.61$ to $8\sqrt{2}\,r^3 \approx 55.58$ at the square and rises again, never reaching 78.61.
- A quick sanity check: a $\sqrt{2} \times \sqrt{2}$ square in a unit circle has perimeter $4\sqrt{2} \approx 5.66$, whereas a 1.9×0.62 rectangle (nearly a diameter) has perimeter about 5.04.

Common slips

- Saying the maximum for $n = 3$ is $16r^3$ "at $\theta = 0$". That is a line segment, not a rectangle, so the maximum does not exist; you must say so and justify the strict inequality.
- Finding the stationary point $\theta = \frac{\pi}{4}$ for $n = 3$ and calling it a maximum without checking its nature.

Beyond STEP

The $n = 3$ case is a first taste of the difference between a *supremum* and a *maximum*: a continuous function on an open interval need not attain its bounds, which is exactly why the extreme value theorem insists on a closed, bounded interval. The switch at $n = 2$ is an instance of the power-mean inequalities, which compare $\left(\frac{a^p + b^p}{2}\right)^{1/p}$ for different p. Abbott's *Understanding Analysis* treats suprema and the extreme value theorem carefully.

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